1. Submit tender / buyer query

8. Draft memo, flags & tool log

2. Plan & reason (prompt + context)

3. Next action / tool choice

Reviewer Web App

Streamlit · zabuni-ai.streamlit.app

- Tender / Buyer Search
- Risk Board & Flag Cards
- Evidence View (cited OCDS)
- Live Tool-Call Log
- Approve / Reject Panel

9. Review flags

Zabuni AI Agent

LangGraph Orchestrator

- Planner – chooses checks & order
- Investigator – follow-ups & recovery
- Synthesiser – ranks evidence, drafts memo
- MCP Client (langchain-mcp-adapters)
- Memory (LangGraph checkpointer)
- Human Approval Interrupt
- Tool-Call Audit Logging

Open-Weights LLM

- Qwen 3.8 27B via Groq (hosted)
- gpt-oss-120b fallback (open weights)
- Qwen 2.5 7B via Ollama (local)
- Planning & Reasoning
- Memo Drafting

4. MCP tool calls (JSON-RPC)

7. Sourced results

10. Approve / reject + reason (gate)

11. Approved flags only

Final Evaluation Memo

- Approved, cited flags only
- Committee decides award

Human-in-the-Loop

- Procurement Officer / Auditor
- Approve / Reject Each Flag
- Reason Required
- Request Clarification
- Agent Never Awards Tenders

MCP

5. Query OCDS & rules

6. Records + source IDs

MCP Servers

zabuni-mcp (custom · FastMCP)

- search_awards
- compute_price_benchmark
- detect_splitting · detect_noncompetitive_method
- supplier_concentration
- detect_price_anomalies (ML)
- flag_finding · draft_eval_memo
- Resources: ocds://release · rules://ppada

Filesystem MCP (external · official)

- Writes memo exports to outputs/ (sandboxed)

Data & Storage

- Historical OCDS Snapshot (Kenya)
- PPADA 2015 Thresholds
- OCP Red-Flag Rules
- SQLite / DuckDB
- Flags & Reviewer Decisions
- Audit Log (timestamps, I/O)